Beyond Score Prediction: Evidence-First Reasoning for Writing Assessment

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Abstract

Gnomon implements an evidence-first architecture for inspectable writing assessment. A compiled rubric connects model-extracted observations to evidence verification, constrained scoring and human-review workflows. Its supporting data system includes source-aware selection, 44 registered quality checks, task- and image-aware training admission, staged practice pools and conditional psychometric target preparation. We explain exact-span transport, relation-level duplicate-credit control, candidate-bound repair selection and memory-aware training loss, with an executed archived-input replay illustrating the scoring boundary. Separate corpus, preparation and training populations are reconciled. A historical, source-calibrated Task-2 record covers 157 essays and reports criterion MAEs of 0.226–0.319. The contribution is an implemented systems integration and its explicit decision interfaces. This is a retrospective architecture and implementation study, not a new comparative accuracy experiment.

1 Introduction

The design question is not simply whether a model can predict a score. It is whether the system can connect that score to the observations, rubric conditions and transformations that produced it, so a disputed judgement has an inspectable path.

Gnomon separates *observation*, *verification* and *scoring authority*. A model extracts criterion-level evidence; tool and verification components assess it; explicit rules construct criterion scores. Here, *reasoning* means inspectable transformations, not access to a model's hidden cognition.

IELTS Writing uses four criteria per task: Task Achievement (TA) for Task 1 or Task Response (TR) for Task 2, plus Coherence and Cohesion (CC), Lexical Resource (LR), and Grammatical Range and Accuracy (GRA). Five criterion names across tasks do not imply five scores on every essay. General Training Task 1 uses a letter instruction rather than an Academic chart (IELTS, 2023a,b).

Contributions. We describe a shared rubric and evidence-to-decision architecture; source-aware preparation and correction-data interfaces connected to training; and a reconciliation of data and run populations with one historical scoring observation. The contribution is the implemented integration and its inspectable boundaries.

2 Related work

Interpretable, feature-informed scoring and criterion-specific assessment are established research directions. e-rater V.2 and the broader automated-essay-scoring literature provide the context for this work (Attali and Burstein, 2006; Ke and Ng, 2019). Gnomon studies how data governance, rubric definitions, evidence transport and constrained scoring can be connected through inspectable interfaces.

Automated-scoring evaluation must be tied to intended use, with agreement, error, bias and validity evidence considered together (Williamson et al., 2012). Structured model judging also remains vulnerable to position, verbosity and self-enhancement biases; an inspectable format does not eliminate them (Zheng et al., 2023; Liu et al., 2023).

3 System architecture

Figure 1 shows the connected information flow. The inspected *grade_engine* entry calls *eval_harness*, which assembles evidence for *scorer_3layer*. Rubric content supplies both elicitation and interpretation.

We distinguish four statuses throughout: *default* means part of the inspected scoring route; *conditional* requires an explicit mode, callback or admit-

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ted configuration; *historical* describes a retained earlier record; and *prospective* identifies a proposed measurement. None is interchangeable with demonstrated assessment efficacy.

3.1 Declarative rubric and extraction contracts

The rubric specification describes constructs, items, dimensions, graded groups, task variations and evidence requirements. The *compile_rubric* implementation validates the specification before emitting the runtime bank, derives a content-based version and can refresh registered dependent pins. The bank supplies both prompt construction and scoring: an item is a shared definition with downstream consumers, not an independently maintained passage of prompt prose. Default content and explicitly included additions remain distinct.

Task metadata informs routing. The extraction seam accepts text and, where applicable, an image; the builder supports task instructions and criterion-specific evidence schemas. Extracted observations are structured around locatable quotations and rubric items rather than a direct essay-band prediction. End-to-end propagation of complete task context through every configured extraction path is not established by this source review.

The extraction contract restricts model output to observations, while scoring rules and calibration references reside in their own components. This is an information boundary, not a claim that all upstream selection is label-free.

3.2 Verification and constrained judgement

The shared scoring consumer combines model-derived item verdicts with deterministic tool signals. It retains distinctions among supported, partially supported, unsupported and unmeasured evidence. Unmeasured evidence and failed presence checks are distinct. In R1, rejection of the legacy flow quotation leaves its presence-rule item UNMET, while the chain and pivot channels return NO_EVIDENCE and are excluded from dimension combination. Zero admissible facts therefore does not automatically mean an unmeasured item.

The default scoring path maps item evidence to criterion dimensions, accounts for graded-ladder structure and applies limiting-dimension aggregation with non-compensatory constraints. A dimension at the top of its measured ladder supplies a lower bound rather than automatically limiting a deeper dimension. Configured psychometric or adopted magnitude-estimation paths can change the base estimate; neither is a universal default. The Rasch branch requires *SCORER_MATH=rasch* and calls *rasch_band* with credited evidence. The complete bank and calibration recipes are not distributed.

For a populated criterion, the order is dimension credit and ladder accounting, limiting aggregation, optional base-estimate substitution, corroborated floor/cap application, half-band rounding, then configured low-tier scale and high-tier uncertainty hooks. The evaluation consumer applies deterministic guardrails after scoring; the engine's decision and human-release boundaries remain downstream. The inspected scale and high-tier hooks are disabled/identity by configuration. An empty dimension set takes the earlier fallback in Table 1, rather than traversing the populated combine.

The retained trace lets a reviewer ask separately whether a quotation exists, whether its interpretation is sound, and which scoring condition it affected.

3.3 Abstention, cross-checks and correction

Disagreement checks can withhold a score when estimates conflict. A single estimate provides no agreement-based confidence evidence, and correlated repeated calls are not independent raters. Evaluation must therefore report coverage alongside error among accepted cases.

The wider engine also contains judge and correction branches. These can cross-check signals, regenerate flagged evidence and rescore it through the shared consumer. Production automatic checks are deterministic-only; model-backed correction and the provenance-bound teaching overlay require explicit modes, callbacks and context. The selected mode determines which branch a report traverses.

Evidence-oriented delivery connects to teacher review, release, learner access and challenge. An internal verification reason is not automatically good teaching: the learner artifact must translate it into a clear diagnosis and manageable action. Workflow source coverage and helpfulness are assessed separately.

3.4 Exact-span transport and relation identity

Supported extraction items can use sentence locators and short head/tail anchors instead of re-

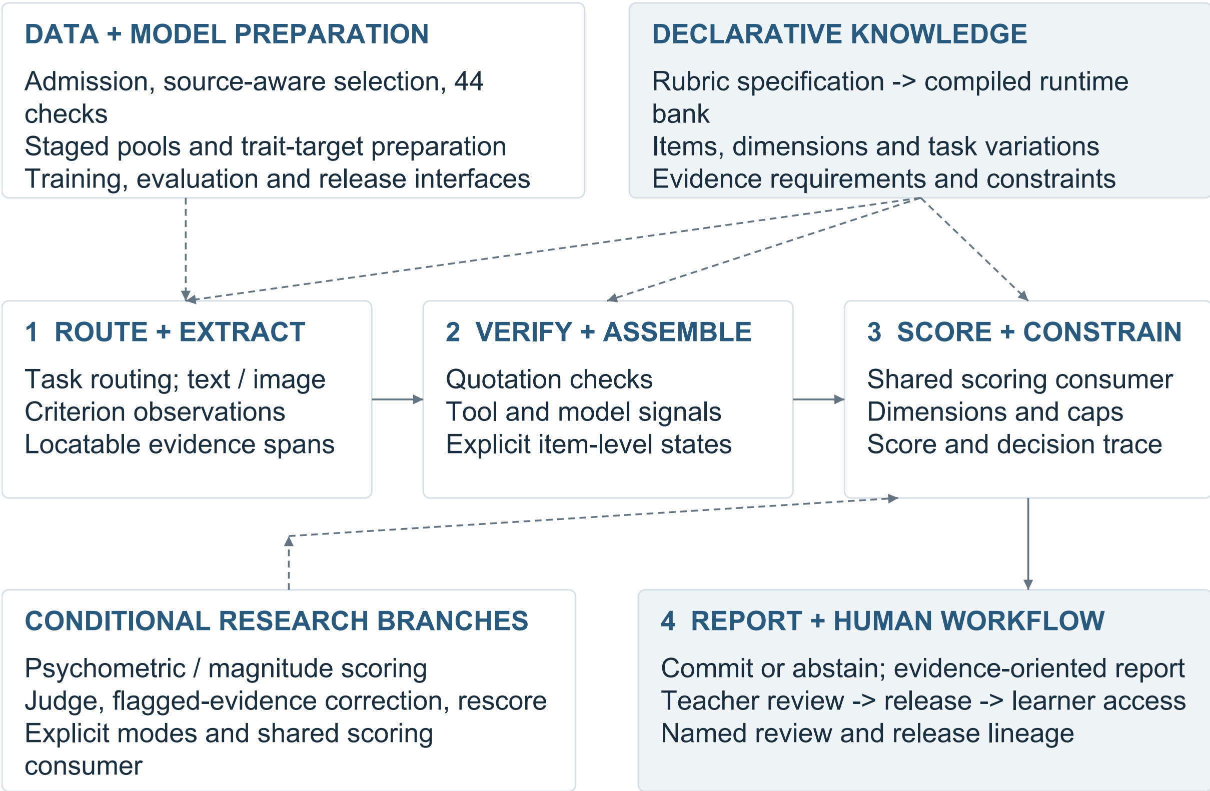

Figure 1: The central assessment path and supporting systems. Dashed links identify configuration or conditional branches; human release remains a separate authority step.

producing a long quotation. The resolver checks the permitted structure, contiguous sentence range, unique matches and ordering, then slices the original response. It does not repair spelling or rewrite the learner's text. Invalid transport rejects the emission or omits the affected fact with an issue, according to the caller's mode.

The relation verifier works above individual quotations. It checks endpoint grounding, cardinality, permitted item memberships, resolution state, scope and agreement among repeated views. Conflicting identities or inconsistent copies invalidate the relation. An essay-local index retains valid views while assigning a resolved relation's positive credit only to its first eligible item in the immutable rubric order. Repeated presentation of the same relation therefore does not create repeated credit. This is mechanical identity accounting, not a judgement that the relation is semantically correct.

These mechanisms distinguish transport identity, evidence identity and semantic interpretation. Exact location supports inspection; it does not prove the interpretation. This creates separate failure boundaries for the model, verifier and scorer, allowing a disputed judgement to be traced without substituting an untracked second opinion.

3.5 Worked extraction-to-score replay

Trace R1 executes the current shared verification, tool, derived-item, scoring and guardrail consumers on the first existing CC cache record, selected before inspecting its result. The archived response and extraction are bound to file, row and text hashes; the replay records the runtime-bank and owner hashes. It uses the heuristic configuration with no model call, new extraction or provider access.

Table 2 presents selected trace observations, not an exhaustive scoring derivation. The dimension positions and logged base of 5.30 also depend on other verified facts in the archived row; the linker examples alone do not determine 5.5. The complete essay and private bank weights are not supplied for independent numerical reproduction.

The original cached input is unchanged. As a separate negative control, replacing the only retained entity-carry-over quotation with absent text changes that item's verdict from MET to UNMET.

Evidence state	Implemented consequence
Supported / unsupported	MET and UNMET are measured states. Credit follows the item's role; a measured flaw can activate a configured cap. An unsupported positive claim is not automatically a flaw.
Partially supported	Graduated items can receive intermediate credit. Neutral and prerequisite items do not receive positive partial credit; unmet measured prerequisites can prevent a higher rung from earning credit.
Unmeasured / unavailable	NO_EVIDENCE, omitted tool signals and unmeasured items do not testify as measured failures. A dimension with no measured item is excluded from combination.
No measured dimensions	The scoring component returns its configured floor, then applies its configured scale/uncertainty hooks. In the inspected CC configuration the result is 4.0, not an automatic abstention. This is distinct from having no score estimate.
Missing task context	Task-scoped checks can be skipped when identity is unavailable. A grounded quotation alone cannot certify task fulfilment. Task-aware admission and upstream input controls are separate contracts.
Decision without estimates	The disagreement layer abstains when its estimate list is empty. With one estimate it commits that estimate; this supplies no independent agreement evidence.

Table 1: Evidence states are not interchangeable. These are source-bound component semantics, not a claim that every route enforces a universal missing-context refusal.

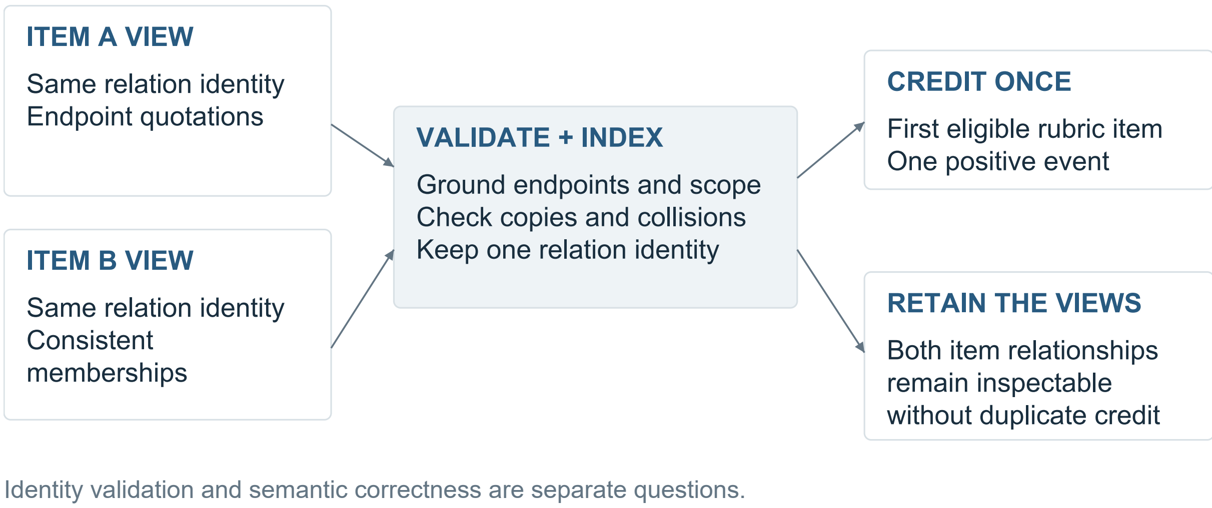

Figure 2: Schematic relation-level accounting. Several item views can refer to one validated relation; scoring credits it once. No learner text or private rubric item is shown.

This tests the grounding boundary, not whether the original entity interpretation was correct. The archived row has no task identity, so task-scoped completeness is not tested; R1 is a compatibility replay, not a claim of current extraction quality or a full learner workflow. Its 5.5 is one computed criterion score, not an MAE.

4 Data and learning infrastructure

4.1 Quality controls with explicit scope

The *run_full_qa_gate* registry contains 44 distinct checks across seven groups. Figure 3 shows that implemented surface. It describes coverage, not a current pass rate.

Controls address exact and normalized duplication, near-duplicate detection, cross-split overlap, missing or malformed fields, source-label trust, distribution coverage and training-readiness conditions. The implementation distinguishes reporting, review requirements and selected repair actions. The training sub-gate selects eight of the registered checks; it is not equivalent to running the entire quality surface.

A central selection API applies source, task, role, quarantine and evaluation-exclusion conditions, with stratification for downstream pools. Asset readiness is closed-world: discovered stores must have an assigned role and a declared route to

Boundary	Observed R1 trace
Source and proposal	The response contains “However” and “On the other hand”. The cached model proposes these as linker observations; the discarded “By the way” proposal is not positive evidence.
Grounding and rule	Two retained linker quotations satisfy the configured presence rule: the attempted-cohesion item is MET. A concatenated, noncontiguous quotation proposed for an overlinking flaw contributes zero grounded facts; that flaw rule is UNMET.
Transport rejection	A legacy raw quotation targets a now anchor-only flow item. The resolver records ANCHOR DROP; zero admissible facts leave this presence-rule item UNMET. Relation identifiers are absent in this legacy record; relation-level duplicate credit is not exercised.
Dimension consequence	The resulting coherence position is 5.0; cohesion is 7.0. Cohesion reaches its measured ladder top and becomes a lower-bound observation, while coherence remains limiting.
Score construction	The limiting-weighted base is 5.30. No cap or guardrail changes this case; half-band rounding produces 5.5. Passing this single estimate to the decision component returns COMMIT. This is not a human-approved release.

Table 2: Selected observations from an executed archived-input replay, not an exhaustive scoring derivation or accuracy test. The trace does not establish that the model’s semantic interpretations are correct.

a consumer or explicit deferment. A declared route is not proof of actual consumption; portions of the graph orchestration remain scaffolds.

Practice-pool preparation stages all five criterion outputs, validates them using the intended reader, preserves previous versions and publishes a manifest last. Replacement is atomic per file. It is not a demonstrated cross-file transaction or crash-consistent whole-set commit.

4.2 Documented asset and run scale

The September inspection reconciled the inventory, parsed the two prepared SFT files and inspected a separate historical training manifest. The counts below have different units and overlap; they are not an additive evaluation population.

The current prepared files are not retrospectively assigned to the earlier checkpoint. These records describe separate assets and a run, not one evaluated population.

4.3 Admission and supervision construction

Training admission checks identity, usable text, task/criterion compatibility, held-out membership, quarantine and selected quality flags. Academic Task-1 achievement examples require a local image through the shared validity owner; specified score-label patterns are rejected from completions. Named reasons make exclusions countable. Label-agnostic teaching can skip label-distribution flags while retaining text-quality and integrity checks. Retired per-essay AI-detection scores are not active authorship gates.

A contrastive assembler uses source-score quantiles to select high/low pairs, then emits paired texts with criterion and dimension identifiers but without numeric scores. Selection is label-dependent; only the emitted representation omits absolute scores. A separate weighted absolute-target assembly is maintained as an optional ablation. Supervision format is therefore an inspectable choice rather than an implicit mixture.

Governed library-write tooling records authorisation, snapshots and before/after row counts alongside the mutation. Together with staged pool publication, it provides a lineage record from selected inputs to emitted artifacts.

4.4 Psychometric target preparation

Three measurement concerns are treated separately: targets used during preparation, evidence-to-score calibration, and confidence monitoring.

Trait-target calibration is distinct from the pairwise contrastive assembler in Section 4.3. The data-preparation module’s historical MFRM name does not establish a many-facet rater model. The inspected implementation estimates trait targets through PCM or normalization; a joint rater-severity model remains a separately proposed protocol. A normalization fallback must not be presented as fitted item-response theory.

The training work includes parameter-efficient multimodal adaptation and completion-focused supervision. LoRA and QLoRA provide established methods for this approach, not evidence of a novel training algorithm or this project’s assessment quality (Hu et al., 2022; Dettmers et al., 2023). Private curriculum construction, prompt contents, adapter settings and calibration artefacts remain

Figure 3: Static registration counts independently reconciled against the source list. Groups are organizational categories, not statistically independent controls or observed quality improvements.

Record	Recorded scope	Interpretation
Corpus inventory	101,175 records; 47 source labels	Generated inventory across heterogeneous sources.
Prepared SFT data	10,888 train rows; 578 validation rows	11,466 parsed preparation records.
Earlier 8B run	9,325 examples; 2,134 image-bearing	Separate non-smoke run; curriculum and weighting enabled.

Table 3: Distinct data-engineering and training records. Source labels are categories, not a count of independent corpora. Preparation rows and historical examples are separate populations.

outside the public overview.

4.5 Critique and repair selection

The research loop separates generating behaviour from admitting a learning target. The *raft_select* module groups model rollouts by essay and evaluates a base faithfulness reward; auxiliary judgement diagnostics do not choose the training winner.

For an essay with k passing attempts out of n, the selector keeps a best passing attempt from the mixed region $0 < k < n$. All-pass examples are skipped by this rule; all-fail examples become teacher-rescue candidates. The emitted completion is serialised from the scored extraction, so a caller cannot substitute a different training completion after selection.

Pass predicate and reward. The evaluated object is the structured extraction and its source response. The *verifier_reward* owner requires quote grounding, at least 90% valid fact schemas, no forbidden score-label leakage, sufficient factual coverage for a long response, and passing anti-copy and distinct-evidence checks. When submission-identity arguments are supplied, task/image identity is checked first; legacy callers supplying none skip that contract. A hard-gate failure cannot be rescued by a higher reward.

The ranking reward combines grounding, schema validity, label cleanliness and non-lazy coverage with a small bonus when both keep and discard paths are represented. The copy and distinct-evidence checks are gate-only. Auxiliary judgement agreement is retained diagnostically and does not reorder the training winner. These predicates address transport and specified extraction failures; they do not certify that every grounded interpretation is semantically correct, task-complete or useful to a learner.

The critique path binds the original candidate,

Mechanism	Engineering role	Qualification
Trait-target preparation	Per-trait partial-credit-model estimation where available; prompt-centred normalization fallback.	PCM when fitting is available; normalization otherwise.
Ranking-aware supervision	Continuous trait-target loss combined with adjacent-pair ranking constraints.	Implemented training-loss interface.
Evidence calibration	Conditional Rasch and graded-response modelling, uncertainty estimates and item-fit diagnostics.	Gated research path, not the default scorer.
Scale and source checks	Monotonic score-scale mapping and cross-source item-invariance diagnostics.	Not joint rater-severity estimation.
Confidence monitoring	Prospective decision/outcome joins and confidence-bin summaries.	Separate from grading-accuracy validation.

Table 4: Implemented calibration-related mechanisms and their roles.

issue set and repair. A revision must differ from the attempt, satisfy the same gate and improve the base reward:

$$R(e_{\text{revision}}) - R(e_{\text{attempt}}) > 0.$$

This is an admission rule, not a theorem that reward improvement implies better teaching. STOP responses, invalid bindings, failed revisions and non-positive deltas retain explicit dispositions. A criterion–mechanism cell is keyed by criterion and the extracted mechanism/item identifier, not by free-text similarity. The anchor owner derives matching keys from ordinary first-pass completions; a correction without a matching anchor is held rather than admitted. These anchors preserve first-pass supervision alongside repair behaviour.

The orchestration mixes base data, admitted attempts and checked corrections, controls self-generated weighting and constructs a new training package. Candidate evaluation and adoption are downstream steps. This is verifier-filtered SFT and correction-data engineering; separate policy-optimisation code does not turn a recorded SFT run into completed RLVR.

5 Training systems and resource accounting

5.1 Completion-focused adaptation

The portable trainer resolves model and precision configuration, checks trainable parameter scope and constructs multimodal batches. Completion-focused supervision masks non-target tokens; per-example weights are carried into the loss interface. Curriculum ordering and length grouping are explicit alternatives: phase order versus grouping supplied sequence lengths to reduce padding. The implementation rejects their simultaneous selection rather than claiming both ordering properties.

5.2 Chunked vocabulary projection

The weighted loss backend projects completion positions through a frozen language-model head in vocabulary chunks. Streaming softmax statistics support full-vocabulary cross-entropy and an entropy term without materialising every sequence position's full logits at once. Backward computation accumulates hidden-state gradients over the chunks; the head must remain frozen.

For a logical optimiser window, the implemented objective is

$$\mathcal{L} = \mathrm{WM}_w(\overline{\mathrm{CE}}) - \beta\, \overline{H}_{\text{completion}},$$

where WM_w is the weighted mean across examples, normalised by the sum of their weights; $\overline{\mathrm{CE}}_i$ averages the completion-token cross-entropy of example i. The entropy mean covers all completion tokens in the window. Common window denominators preserve the intended weighting across microbatches. A materialised reference supplies an arithmetic comparison target. Floating-point reduction order can differ; no new kernel-performance measurement is reported.

5.3 Useful-work denominators

The trainer separately records input tokens, non-padding tokens and completion-loss tokens from windows whose optimiser steps actually applied. These describe batch throughput, useful-token throughput and supervised-target throughput. Keeping them distinct prevents padding from being credited as the same unit of learning work.

Admission, checkpoint evaluation, packaging and serving interfaces connect training to release decisions. Training completion, model selection and serving admission remain separate states. The historical 8B manifest records one execution, not a claim that later trainer components ran in that experiment.

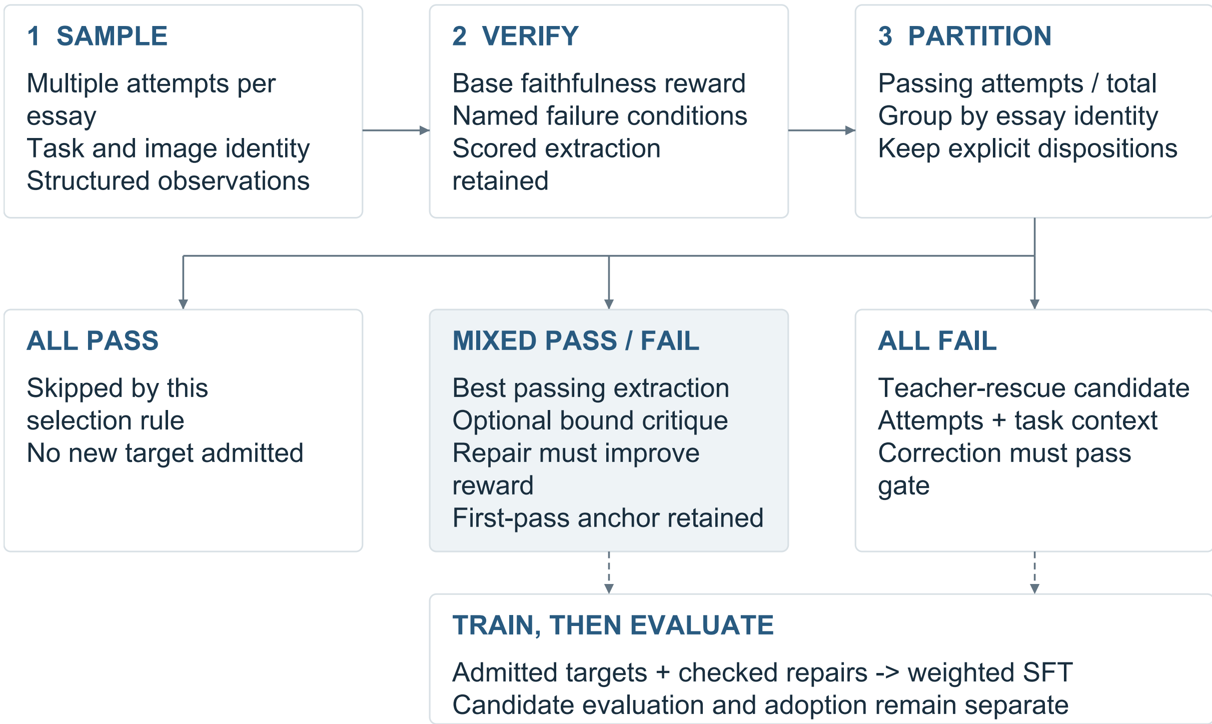

Figure 4: Implemented selection and training interfaces. This is a configured research loop, not a record of a completed production continual-learning campaign. Passing refers to the specified verifier conditions.

5.4 Earlier encoder engineering

Historical ModernBERT work includes per-layer low-rank adapters, fused/separate projection handling and inference-time merging. A DirectML-compatible AdamW path preserves optimiser state and bias correction using supported updates. These are provenance for earlier engineering, not dependencies of the current extractor or evidence of its efficacy.

6 Evaluation scope and observations

6.1 Study method

The object of study is the integrated assessment pipeline: its data contracts, evidence transformations and decision interfaces. The scope is not general-purpose benchmarking of the underlying provider models. Evidence comprises implementation inspection, record reconciliation and reproduction of the displayed aggregate figures.

The September 2026 review used existing registries and boards to locate source owners and direct consumers, then reconciled the cited data and run records. An initial 46-owner pass was followed by targeted inspection of admission, relation handling, correction selection and training loss. Private claim-to-owner mappings retain source hashes. Default paths, conditional implementations, historical observations and prospective studies are separate evidence classes.

The accompanying *build_figures.py* defines three implementation schematics and two aggregate plots. Its numerical inputs are also exported as *figure_data.json*. Running *python build_figures.py* with ReportLab and Arial/Arial Bold redraws those figures and checks the aggregate arithmetic; the schematic definitions are not measurements. This reproduces the displayed figures, not the original assessment experiment. R1's separate replay script invokes the existing scoring owners on its pinned archived input and emits the resulting trace and checks. It requires the private input and matching runtime, unlike the aggregate figure generator.

6.2 Historical development record

Earlier Task-2 development records report criterion MAEs of TR 0.319, CC 0.268, LR 0.248 and GRA 0.226 on 157 essays, with source-specific calibration. Figure 5 preserves that single, scoped record. It is not a present-engine benchmark or an independently reproduced held-out result.

Complete prediction/reference rows, calibration membership and immutable model configuration have not been reconstructed for this report. Conse-

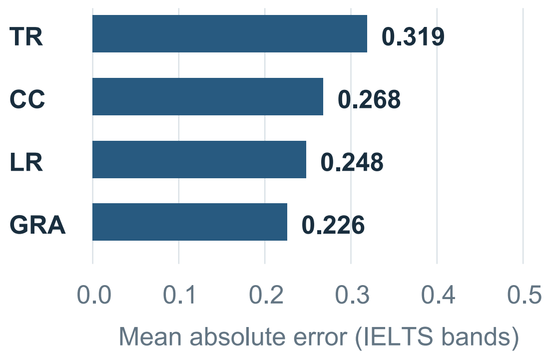

Figure 5: Historical Task-2 MAE on 157 essays with source-specific calibration. Reported correlations: TR 0.941, CC 0.915, LR 0.919 and GRA 0.921. These aggregates are not a current-engine or independently reproduced held-out benchmark.

quently, these observations motivate testing rather than establish generalization.

7 Discussion

Gnomon's engineering proposition is the integration: governed data, a compiled rubric, constrained evidence extraction, traceable scoring and explicit correction interfaces. The intended benefit is locating and correcting decision errors. A matched comparison with a direct call to the same model is the appropriate test of the integration's incremental accuracy, usefulness and cost.

The decisive next evaluation should use previously unused, input-complete essays, fixed references and calibration outside the test cohort. Compare the present pipeline with a direct call to the same model under disclosed resource budgets. Retain every prediction, refusal and exclusion; report criterion errors, uncertainty, coverage, bias and total cost. Teaching quality needs an additional blinded comparison of complete feedback, followed by learner comprehension and fresh delayed transfer. These prospective tests assess outcomes beyond implementation.

8 Conclusion

Gnomon implements a connected assessment architecture in which rubric definitions, extracted observations, verification conditions and score construction share explicit interfaces. Its supporting infrastructure extends that discipline to data admission, correction selection and training-resource accounting. The contribution is an inspectable path from source data and evidence to a decision, with defined points for correction and human authority. Matched studies will test its practical value.

Limitations

The evidence is retrospective and configuration-specific. Implementation inspection and record reconciliation establish the scope of the described mechanisms; current end-to-end accuracy, operating scale and teaching effectiveness require separate measurement. Direct-call advantage, teacher preference and learner benefit are questions for the comparisons specified above. Current dataset quality requires a fresh, version-bound assessment.

Private datasets, rubric contents, prompts and calibration recipes are outside the public release. The supplied aggregates and figure code reproduce the displayed plots; independent assessment-run reproduction additionally requires the restricted inputs, row-level predictions and exact configurations. The historical result is interpreted only within its recorded population and calibration.

Ethical considerations and AI assistance

The intended application is research into formative writing support, not autonomous high-stakes grading or certification. Misinterpreted evidence and unhelpful feedback can mislead learners; source, task and language-background imbalance can produce unequal errors. Human review is a proposed safeguard, not proof those risks have been removed.

The report presents aggregates and short, non-identifying phrases from R1, without distributing the learner response, source record or institutional identifiers. Source licences, consent conditions, examiner qualifications and ethics approvals are outside this inspection and must be established for the intended data sharing, deployment or human study. No official IELTS endorsement is claimed.

Claude and OpenAI Codex assisted implementation, drafting, reference checking and figure-generation code. Three figures are programmatic implementation schematics; two plot disclosed aggregates. AI consultation and tool checks are not independent human peer review. The author retains responsibility for final claims, citations and permitted data use.

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